

2025-2026 Vize Sınavı

1) Aşağıda verilmiş olan sayı sistemi dönüşümlerini yapınız. (20 puan)

a) $(A8)_{16} = 10 \times 16 + 8 \times 1 = 168$

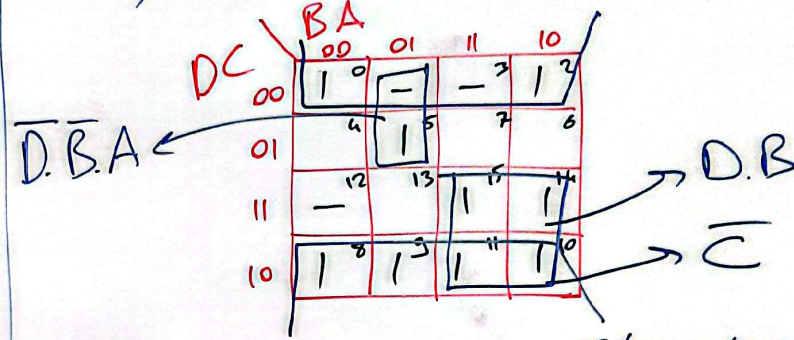
b) $(\underline{110101})_2 = (65)_8$

c) $63 = (?)_2$ $64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1$
 $0(111111)_2$

d) $(47)_8 = (27)_{16}$

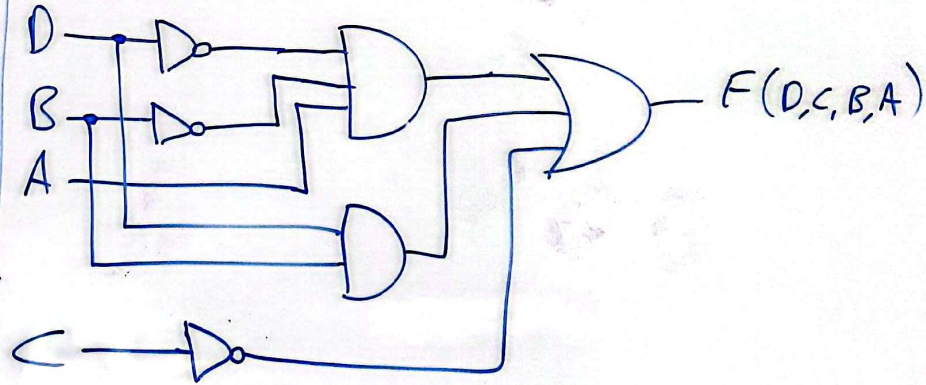
$(\underline{00100} \ \underline{111})_2$
 $2 \quad 7$

2.) $F(D,C,B,A) = \sum m(0,2,5,8,9,10,11,14,15) + \sum \phi(1,3,12)$ Mantık ifadesinin en sade halini Karnaugh Haritasını kullanarak bulunuz. (20 puan)

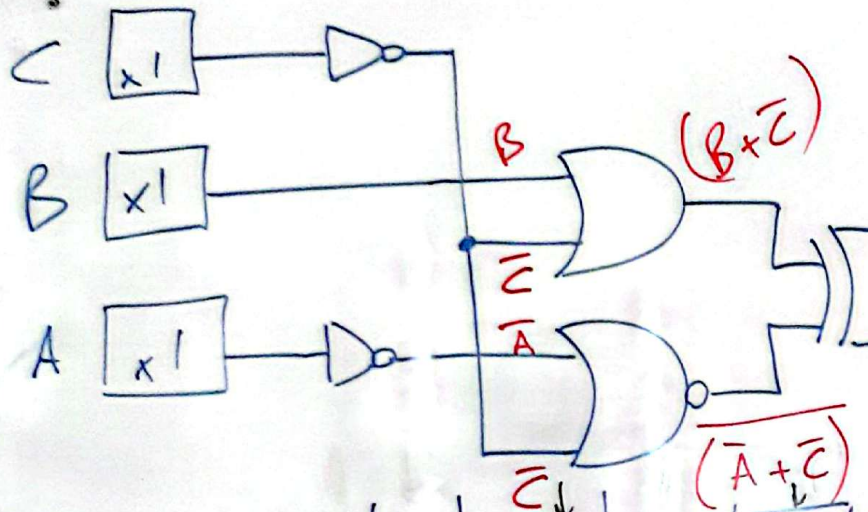


$$F(D,C,B,A) = \overline{D} \cdot \overline{B} \cdot A + D \cdot B + \overline{C}$$

b) En sade halini sadece VE/VEYA/DEĞİL kapıları kullanarak devre şemasını çizin. (30 puan)



3-) Aşağıdaki bileşik mantık devresini analiz edip maksiterim gösterimini (Π_M) yapınız (20 puan)



$$Q(A, B, C) = \Pi_M(1, 7).$$

$$\overline{(B+C)} \cdot \overline{(\bar{A}+C)} + (B+C) \cdot (\bar{A}+C) = \bar{B} \cdot C \cdot A \cdot C + (\bar{A} \cdot B) + \bar{C}$$

A	B	C	$\bar{A}$	$\bar{C}$	$B+C$	$\bar{A}+C$	$\bar{A}+\bar{C}$	$Q(A, B, C)$
0	0	0	1	1	1	1	0	1
0	0	1	1	0	0	1	0	0
0	1	0	1	1	1	1	0	1
0	1	1	1	0	1	1	0	1
1	0	0	0	1	1	1	0	1
1	0	1	0	0	0	0	1	1
1	1	0	0	1	1	1	0	1
1	1	1	0	0	1	0	1	0

→ 1

→ 7

II. ypl

$$\bar{B} \cdot C \cdot A \cdot C + (\bar{A} \cdot B) + \bar{C}$$

$$1 \cdot 0 \cdot 1 \cdot 0 + 0 \cdot 1$$

A \ B \ C	0	1
00	1 ⁰	0 ¹
01	1 ²	1 ³
11	1 ⁶	0 ⁷
10	1 ⁴	1 ⁵